# Nordic Flow: fix the allocation decision first, with one planner-approved service that lowers total logistics cost

Executive summary for the Board of Company A: answers to the questions the case asks

**8 weeks**
from approval to a go / no-go decision

**≥ 5%**
lower relevant logistics cost: pilot gate [assumption]

**~3–6%**
saving needed to break even in year 1 [assumption]

**0**
ERP or system replaced; planner approves every plan

| Question | Our answer |
|---|---|
| **Which problem?** | Goods allocation: which source and existing warehouse should serve each order. It sits where fulfilment, manual planning and fragmented systems meet. |
| **Who experiences it?** | Planners and warehouse teams in every market; customers feel it as late or split orders. |
| **Why does it matter?** | Margins are thin, so cost-to-serve is a real lever; outside studies show savings from ~5% (modelled) to 10–15% (realised) [cited]. |
| **What do we propose?** | A planner-approved optimisation service on top of existing ERP, WMS and TMS. Nothing is replaced. |
| **MVP scope?** | Sweden, one region: 2 suppliers, 3 warehouses, 20 SKUs, weekly run; 8 weeks to a go / no-go [assumption]. |
| **Where does AI help?** | Data mapping, plain-language explanations, learning from overrides. The maths is optimisation; a person approves. |
| **How does it scale?** | Same engine, local data and rules: Sweden, then Denmark, then Norway, each gated by service and savings. |

Sources and tags are on slides 3, 8 and 9. [assumption] = to be replaced with Company A data in Week 1.

# Goods allocation leads when impact, speed and data readiness are weighed together

Five opportunities scored 1–5 by the team against case evidence; scores are judgement, not company data

| Criterion (weight) | A Goods allocation | B Manual order processing | C Store standardisation | D Systems replacement | E Delivery visibility to site |
|---|---|---|---|---|---|
| **Business impact (30%)** | 4 | 3 | 3 | 5 | 5 |
| **Fit across DK / NO / SE (25%)** | 4 | 3 | 2 | 5 | 3 |
| **Time to market (25%)** | 4 | 4 | 2 | 1 | 3 |
| **Data readiness (20%)** | 3 | 3 | 2 | 1 | 2 |
| **Weighted score** | 3.80 | 3.25 | 2.30 | 3.20 | 3.40 |

### Why A

An internal decision: no new driver, store or customer-facing change.

Uses data the company already holds, so a CSV-based pilot can run in weeks.

### Robustness check

If impact is weighted 50%, E ties A at about 3.86 each.

So E is our Phase 2: it reuses the same order and warehouse data.

### Why not the others now

B: helps mainly Denmark; most savings claims come from vendors.

C: needs broad discovery in each store.

D: multi-year, high risk.

Weights: impact 30%, fit across three markets 25%, time to market 25%, data readiness 20%. Case pp. 9–12 for the four challenges; delivery visibility added by the team.

# The allocation lever is plausible and modest, so Week 1 measures it before we commit

Separating what the case confirms from what we must verify

## Confirmed in the case

- Hundreds of stores in Denmark, Norway and Sweden, each market with its own systems.
- Fulfilment and fragmented systems affect efficiency and reliability.
- Impact and urgency differ by market.

## Hypotheses to test in Week 1

- Planners optimise one leg at a time, not total cost.
- Rates, stock and capacity sit in separate local systems.
- Stock records may be wrong, so plans send goods to warehouses that do not have them.

## Outside evidence (other settings)

- 4.8% lower delivered cost, building-products maker; modelled [cited]
- 10–15% logistics saving, 150+ warehouse network; realised [cited, consultancy]
- 65% of stock records inaccurate; profit down >10%, one US retailer [cited]
- Byggmax restructured logistics "to increase control over flows" [cited]

**Who feels it:** planners and warehouse teams (manual re-planning), branch staff (stock not where promised), customers (late or split orders). **Why now:** thin margins (Byggmax EBITA 5.9% [cited]) make cost-to-serve a lever.

**Reading:** gains of a few percent are plausible, not guaranteed. We therefore gate the pilot on a measured cost gap, not on a promised saving.

Sources: Case brief pp. 8–11; David et al., Interfaces 45(3), 2015; McKinsey distribution-network case; DeHoratius & Raman, Management Science 54(4), 2008; Byggmax Annual Report 2025, p. 6.

# A focused pilot: one Swedish region, three warehouses, 20 SKUs, one weekly planning run

Scope is the smallest network that still contains a real routing choice; all sizes are assumptions Company A must confirm

| 2 | 3 | 20 | 1 |
|---|---|---|---|
| suppliers | existing warehouses | palletised SKUs | weekly planning run |

## In scope

- Allocate inbound supply and confirmed orders across existing warehouses
- Compare the current allocation with a feasible lower-cost plan, with cost by leg
- Planner approves; every override is logged

## Out of scope

- New warehouses or site selection
- Vehicle routing (stays in the current TMS)
- Demand forecasting, production planning
- ERP replacement; other countries

**Why Sweden first:** the case ties Sweden to fulfilment impact. It is a discovery candidate, not a commitment: if Week 1 data is better in Denmark or Norway, the pilot market changes.

Sizes are [assumption]. Byggföretagen forecast: building investment +6% in total 2026–27, a fragile recovery [cited, autumn 2026 update]; case p. 10 lists delivery and fulfilment as Sweden's priority.

# The planner stays in control: the service recommends, a person approves, existing systems execute

The recommendation fits today's process instead of replacing it

### 01 Load
Orders, supply, rates and warehouse limits from existing systems

›

### 02 Compare
Current allocation versus a feasible lower-cost plan

›

### 03 Review
Exceptions, service rules and cost by leg

›

### 04 Approve
Planner exports the plan; ERP / WMS / TMS execute it

## Recommendation output (example)

| Supplier / SKU | Warehouse | Destination | Quantity / required date |
|---|---|---|---|
| **S1 / SKU-01** | Warehouse A | Order zone North | 60 pallets / within the week |
| **S2 / SKU-01** | Warehouse B | Order zone South | 40 pallets / within the week |

### Not in the pilot

• ERP replacement • Automatic execution of plans • Replacing planner judgement

**Design principle:** reuse before build. Local systems stay authoritative; we add one thin layer: CSV / API adapters, a validation step, the optimiser and a planner view.

## Data the pilot needs (existing systems)

| Data | Where it lives | Owner |
|---|---|---|
| **Supply and orders** | ERP | Planner |
| **Rates and lanes** | TMS, carrier contracts | Logistics |
| **Capacity and stock** | WMS | Warehouse lead |
| **Service rules** | Local policy | Ops sponsor |

Week 1 checks that each item exists and is usable.

Illustrative output only. Each run stores the input version, solver status, approvals and planner overrides.

# A constrained optimisation engine does the maths; AI helps around it, always with a human approving

Optimisation research, not a black box: every constraint and solver status is visible

## Decision engine

**Minimise** inbound freight + handling + outbound freight + storage fees that change with the plan

**Subject to** demand met, supply limits, warehouse handling limits, SKU eligibility, allowed lanes

**LP** for divisible quantities; **MILP** for whole pallets and discrete commercial rules

**Solver status shown:** optimal, feasible within time limit, or infeasible with a diagnosis (short on supply, capacity or lanes)

Fixed rents already committed are not counted as savings.

## Where AI adds value

**1 Data mapping.** Suggests SKU and unit matches across country systems; a person validates.

**2 Plain-language explanations.** Why a warehouse changed and which constraint binds, so the planner can check quickly.

**3 Learning from overrides (Phase 2).** Planner corrections refine rules over time.

**Technology:** Python service with open-source OR-Tools, EU-hosted, role-based access; scenario store keeps input versions, solver status and approvals for audit.

**Responsible by design:** goods allocation is not an Annex III high-risk use [cited]; no profiling of staff; data stays in EU hosting.

EU AI Act, Regulation (EU) 2024/1689, Art. 6 and Annex III. Google OR-Tools is open source (Apache 2.0).

# A worked example shows where the saving comes from: total cost across both legs

Synthetic data: 1 SKU, 2 source warehouses, 2 destination zones; cost units per pallet; both plans serve the same 100 pallets

| Source | A / North | A / South | B / North | B / South |
|---|---|---|---|---|
| **S1** | 5 | 9 | 11 | 8 |
| **S2** | 6 | 10 | 8 | 5 |

Supply S1 / S2 = 60 / 40. Demand North / South = 60 / 40. Warehouse capacity A / B = 100 / 60.

| Plan | Allocation | Total cost |
|---|---|---|
| **Baseline: preferred warehouse A** | 60 via S1–A–North + 40 via S2–A–South | 700 |
| **Optimised** | 60 via S1–A–North + 40 via S2–B–South | 500 |

**700**

baseline

**500**

lowest feasible cost

### What it shows

- The saving (200 units, 28.6%) comes from seeing both legs together, not from a cheaper single leg.
- Habit fills the preferred warehouse; the optimiser uses the cheaper route that still respects capacity.
- Real gains are expected to be far smaller than 28.6%; see the next slide.

Synthetic example for illustration; it demonstrates the mechanism and is not a forecast for Company A.

# The case to proceed: a saving of only ~3–6% breaks even in year one, and Week 1 measures the real figure

We do not promise ROI; we show what must be true and gate the pilot on it

**Annual saving =** relevant logistics cost in pilot scope × saving rate

**Break-even rate =** (build ≈ €0.2M + run ≈ €0.1M) ÷ relevant cost

| Relevant cost in scope | Saving at 2% | at 5% | at 10% | Break-even |
|---|---|---|---|---|
| **€5M / year** | €0.10M | €0.25M | €0.50M | 6.0% |
| **€10M / year** | €0.20M | €0.50M | €1.00M | 3.0% |

## Reading the table

- Only costs that change with allocation count; committed warehouse rents do not.
- The 5% gate clears break-even when the relevant cost is about €6M or more; Week 1 confirms the base.
- If Week 1 finds a smaller cost base or gap, we stop.

## Pilot acceptance gates

| Measure | Gate (hypothesis) |
|---|---|
| **Relevant logistics cost** | ≥ 5% below a validated feasible baseline, same orders |
| **Planner preparation time** | ≥ 30% lower, full task incl. data checks |
| **Service** | No deterioration in on-time, in-full rate |
| **Feasibility** | Zero hard-constraint violations |

**Scale only when** validated savings cover delivery and operating cost. No revenue uplift is assumed.

All cost inputs are [assumption], replaced with Company A data in Week 1. Outside benchmarks: 4.8% modelled (Interfaces, 2015), 10–15% realised (McKinsey). Labour savings assessed separately.

# One engine, three rollouts: each market gets the same core with local data and rules

The case: one strategic solution does not mean one identical implementation

| Wave | Market | What changes locally | Evidence to collect first |
|---|---|---|---|
| **1** | Sweden | Pilot one region from CSV exports; planner reviews every plan | Rates, stock, service rules, warehouse capacity |
| **2** | Denmark | Reuse the engine; adapt the import to cut manual preparation | Structured B2B e-invoices only 5–10% [cited], so inputs need more cleaning; wholesale EDI sales 39% [cited] |
| **3** | Norway | Validate branch and distribution workflows before connecting | Structured B2B e-invoices 89% [cited]; wholesale EDI sales 27% [cited]; store receiving limits, carrier rules |

### Shared core

Data contract, optimisation rules, audit trail, KPI definitions.

Local teams own rates, service rules and approval rights.

### Employees

Planners co-design the workflow and keep the final say; overrides are logged, not punished. Training in the first week of each wave.

### Customers

No change to ordering. On-time, in-full rate is a gate before each wave, so service cannot quietly worsen.

Eurostat isoc_ec_eseln2 (2024, wholesale, ≥10 employees); Digdir (NO) and Erhvervsstyrelsen (DK) e-invoicing analyses. Rollout order is a proposal, not a finding about local systems.

# Eight weeks to a pilot decision, with the main risks designed out and a clear ask

Indicative plan after sponsor approval and access to usable data

### Week 1

Validate problem and data

Exit: Agreed baseline, scope, service rules

### Weeks 2–4

Build MVP: engine, import, plan view

Exit: Plans pass historical replay

### Weeks 5–6

Controlled trial with local planners

Exit: Planner acceptance, service evidence

### Weeks 7–8

Review economics and readiness

Exit: Scale, revise or stop against gates

| Risk | Mitigation |
|---|---|
| **Stock or rate data is wrong (65% of records in one study)** | Validation step; block invalid runs; planner approves |
| **Inbound freight is paid by suppliers (seen at Byggmax)** | Week 1 confirms who pays; if suppliers, optimise warehouse-to-customer leg and handling only |
| **Weak baseline overstates the gain** | Compare against a validated feasible baseline on the same orders |
| **Low planner adoption** | Co-design; keep overrides; show cost by leg |

## Decision requested

Approve Week 1 Swedish discovery and a capped pilot design.

Name a business owner and a local process owner.

Proceed only if data, a measured cost gap and ownership are confirmed; otherwise iterate or stop.

Stock-record accuracy: DeHoratius & Raman (2008). Plan and sizes are indicative and depend on data access.